



Arduino® UNO Q Build Anything

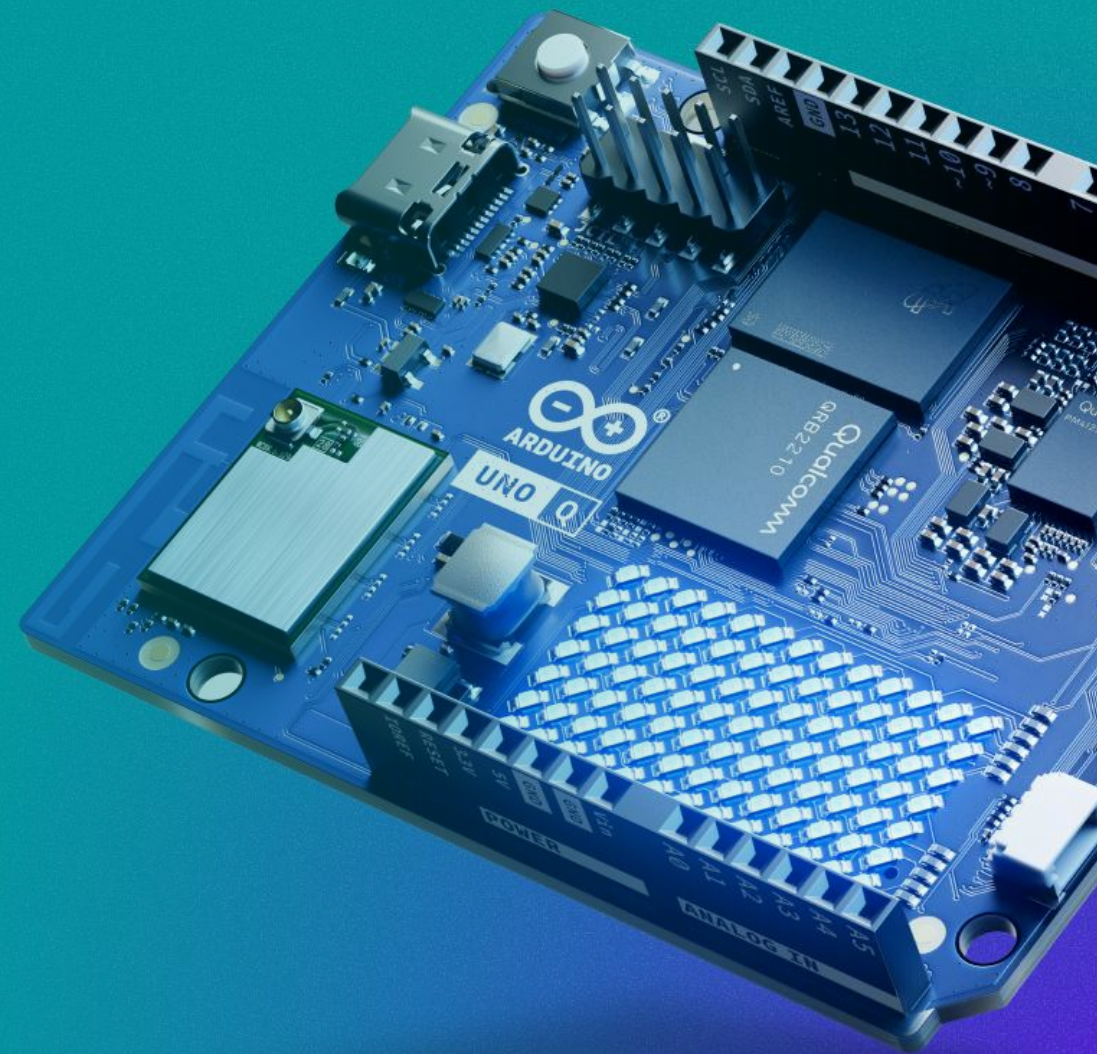
Getting to know the Arduino® UNO Q and
Arduino App Lab platform



What is Arduino?

- Open-source coding & electronics platform
- Founded in 2005
- Tool for designers, artists, hobbyists to get creative with electronics & programming
- Simplified code > accessible for everyone

An Arduino is a circuit board that you program to do specific task(s).



Arduino® UNO Q

Dragonwing™ QRB2210 Microprocessor (MPU) running a full Linux environment (Debian)

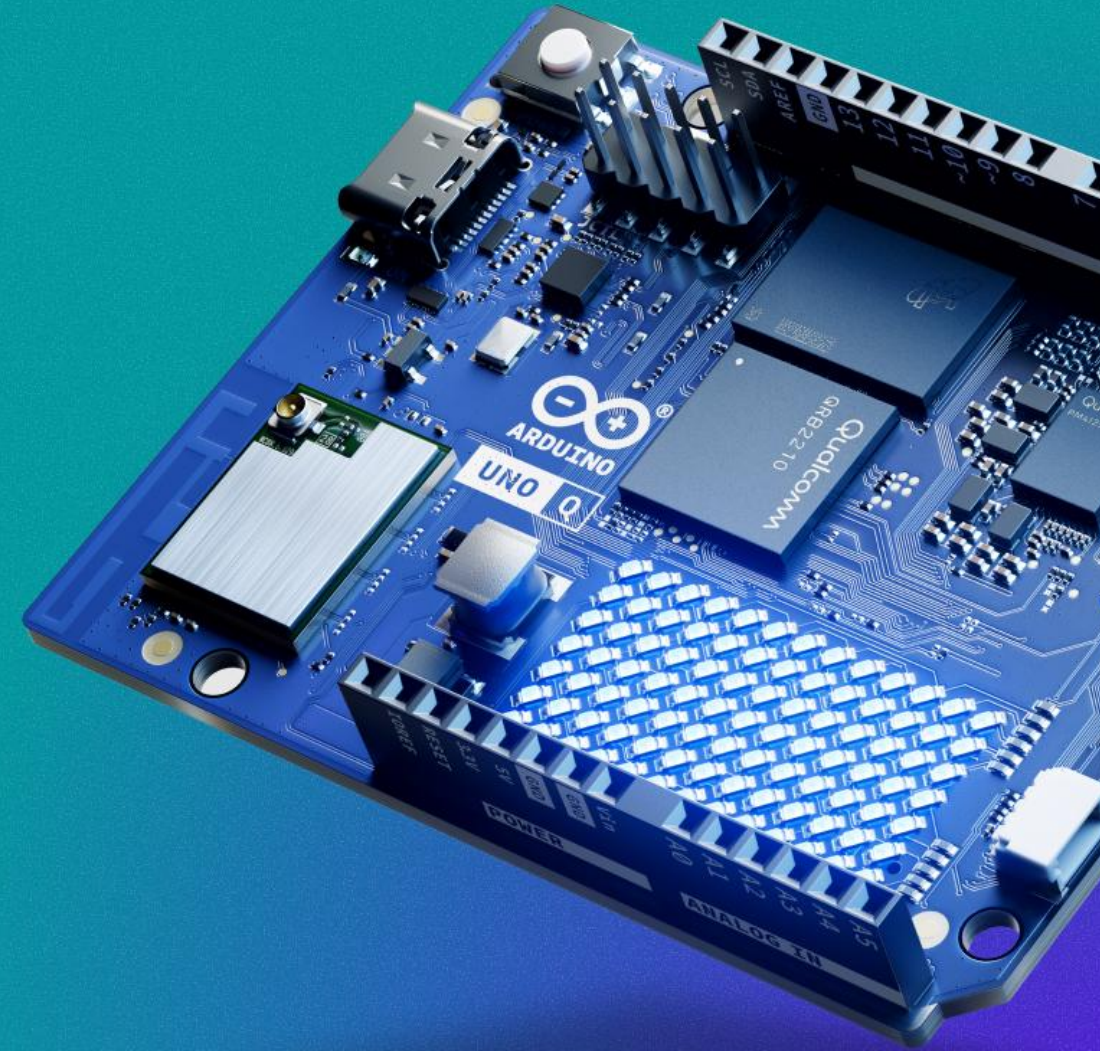
STM32U585 Microcontroller (MCU) running on Zephyr OS

Bridge library that makes them talk

Wi-Fi / BT for connectivity

LED Matrix for feedback & fun graphics

2GB/4GB RAM with **16/32GB eMMC** storage



Arduino App Lab

The IDE for everything

Bricks are modular **building blocks**, used to build **Apps**.

Bricks can for example contain an **AI model** that can run inference on image / audio

They can also host a web server or fetch latest weather data



Launching an App is just a few steps:

1. Pick an example
2. Make modifications
3. Launch the App

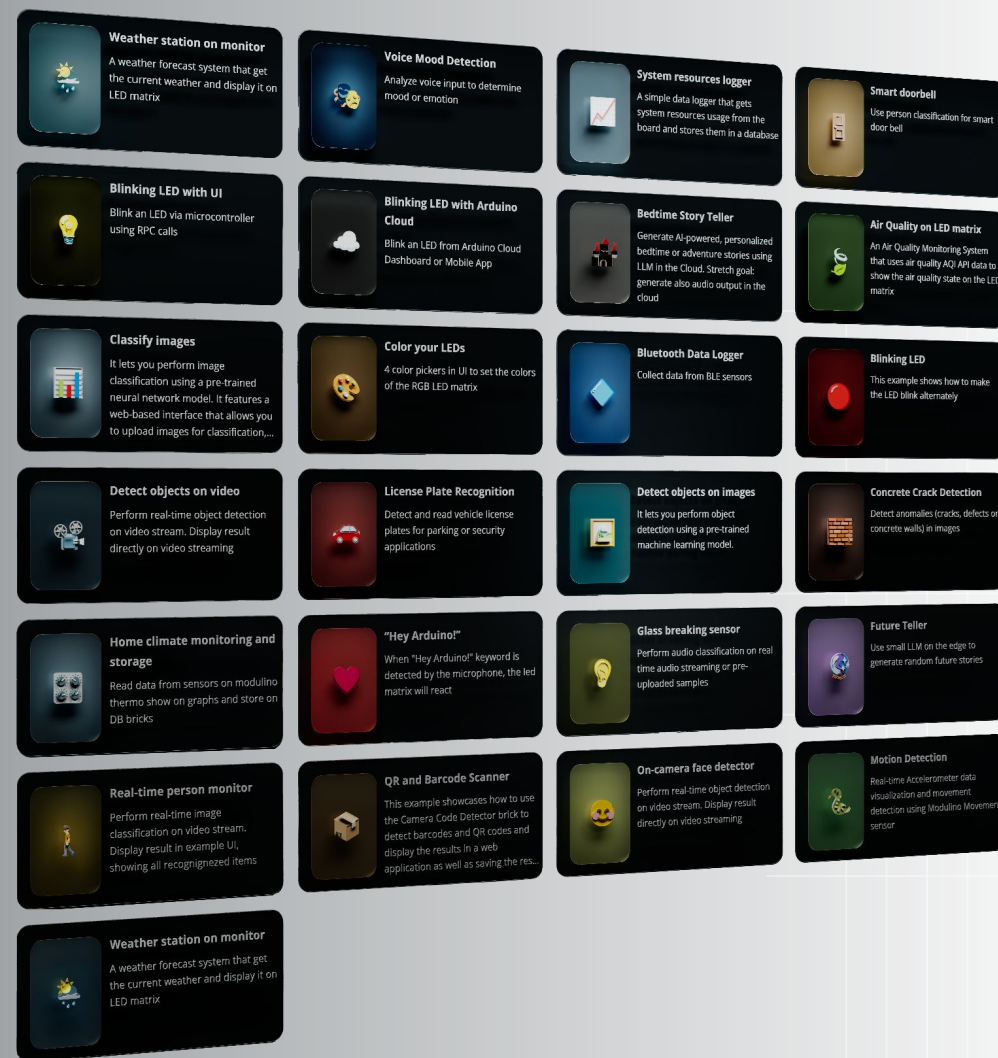
When launching the App, a sketch is uploaded to the MCU, and the MPU will run a Python script, making use of the Bricks.



30+ official examples

Machine Vision
Cloud AI Assistant
Audio Detection
Web Interface

All examples run with generic USB cameras,
microphones & modulinos





Single Board Computer

Debian OS

Display + keyboard + mouse (USB hub)

Arduino App Lab pre-installed

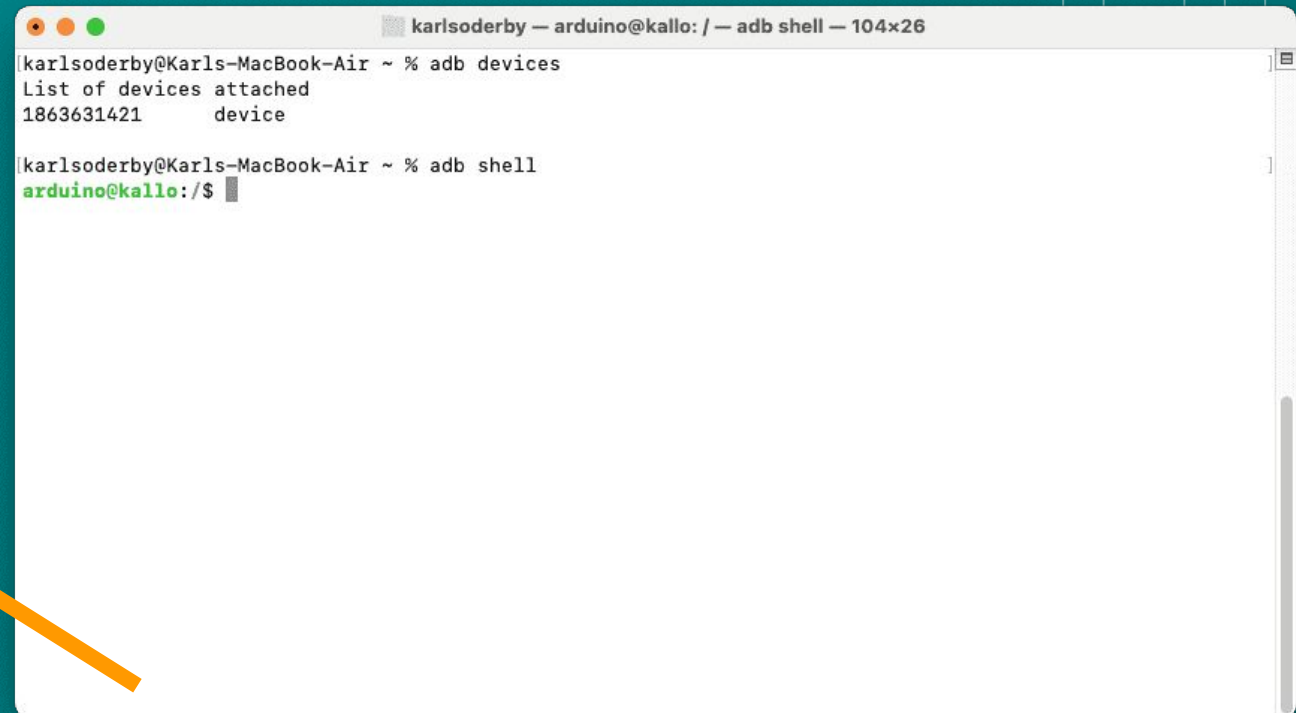
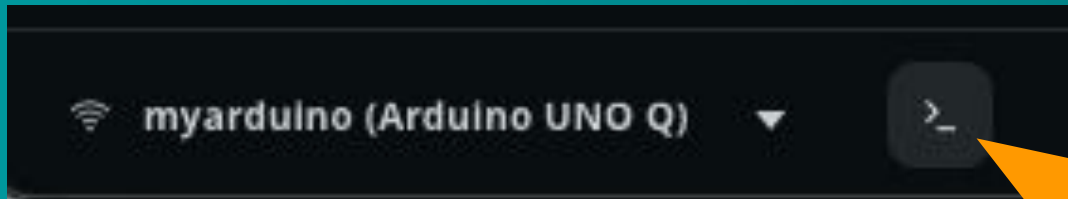




Access your UNO Q via Terminal

1. Install **adb** on your computer
2. or use **SSH**

Terminal is also accessible from the Arduino App Lab.





Agents Setup

1. Install **adb** on your computer
 - a. Mac: **brew install --cask android-platform-tools**
 - b. Windows: **winget install Google.PlatformTools**

adb is a tool that allows you to access the UNO Q over USB-C. It is a requirement for this hackathon.

Depending on your agent permission setting, your agent may be able to install this for you.



What are we doing today?

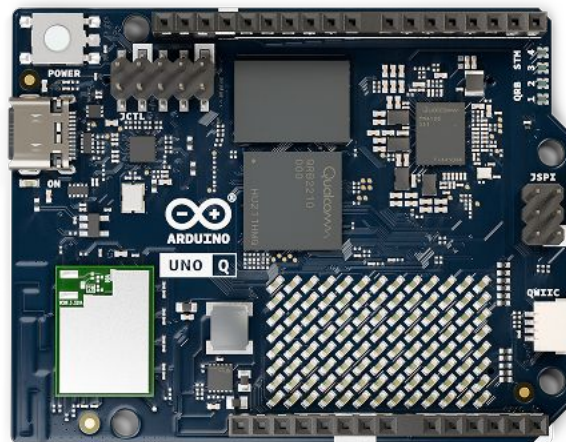
**We are building a
GAME DECK!**



(sorry for the AI slop)



Hardware Overview

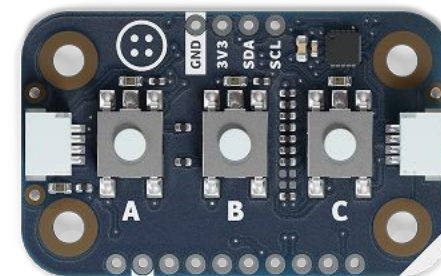


UNO Q 4GB



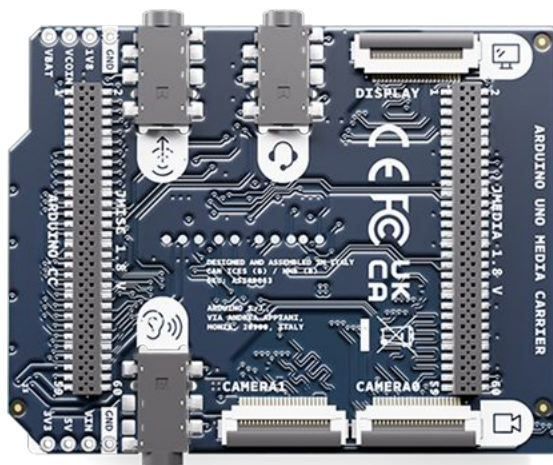
Modulino

Joystick



Modulino

Buttons



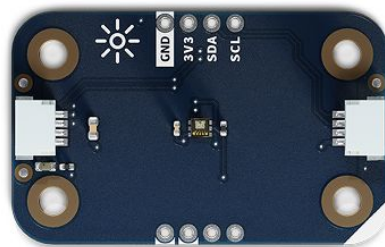
Media Carrier



5" Touch Display (DSI)

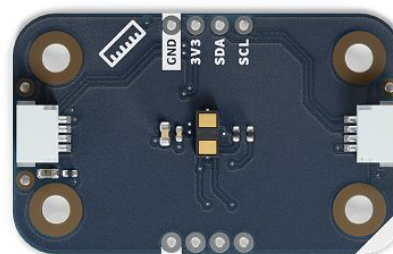


Hardware Overview page 2



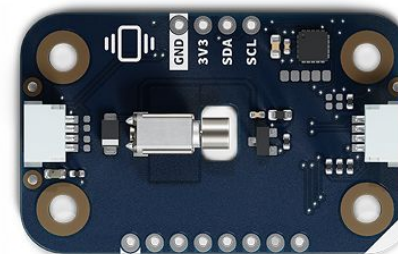
Modulino

Light



Modulino

Distance



Modulino

Vibro



USB-C Cable



Qwiic Cable



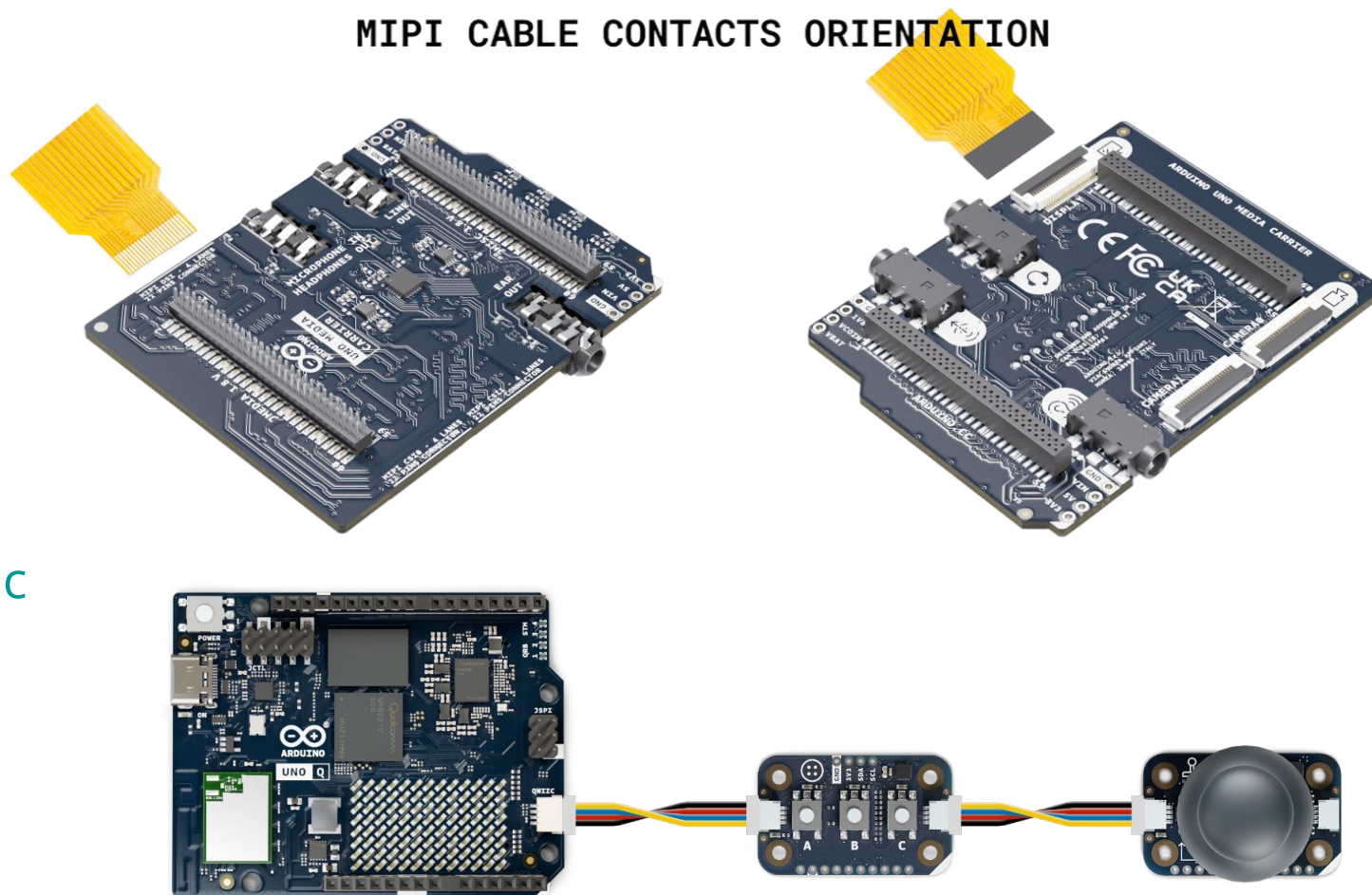
DSI Cable



Hardware Guide

1. Connect UNO Q and Media Carrier (make sure connectors match)
2. Connect display (see picture >>).
3. Connect Modulinos (using the Qwiic Cables)
4. Power the board via USB-C

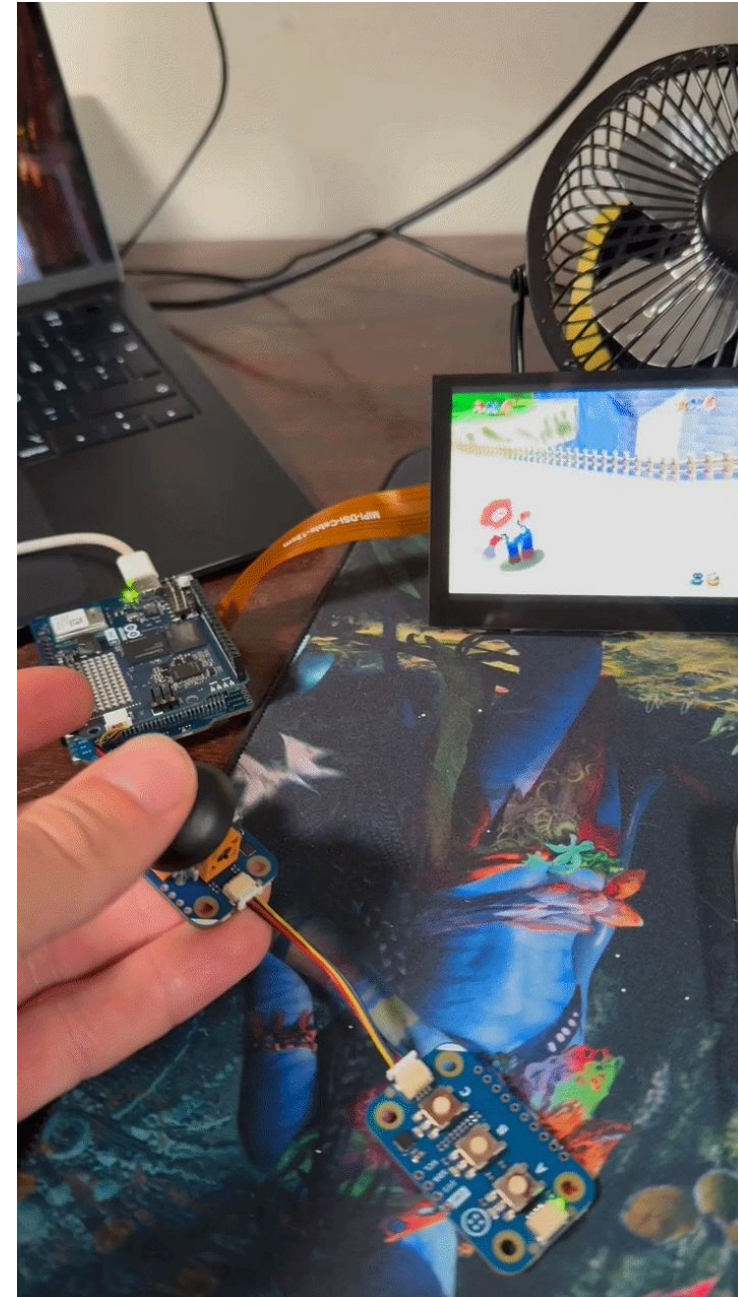
MIPI CABLE CONTACTS ORIENTATION





Final assembly

February, 2026
CONFIDENTIAL © Arduino





First Step

- Download Arduino App Lab
 - <https://www.arduino.cc/en/software/#app-lab-section>
- Grab an Arduino UNO Q
- Grab accessories (Display, cables, modulinos)
- Assemble the hardware
- Clone this repository
 - <https://github.com/karlsoderby/uno-q-modulino-examples>
 - this repository contains several examples, including joystick, HID and everything you need to use Modulinos.



Next Step

Paste this prompt into your agent (cursor, claude code, codex etc etc):

“Install Summer Engine and let's make a game for the Arduino Uno Q. Read github.com/SummerEngine/summer-uno-q, check what I already have installed, install whatever is missing, and log me in.”

This will make it possible to develop and export games directly to the UNO Q.



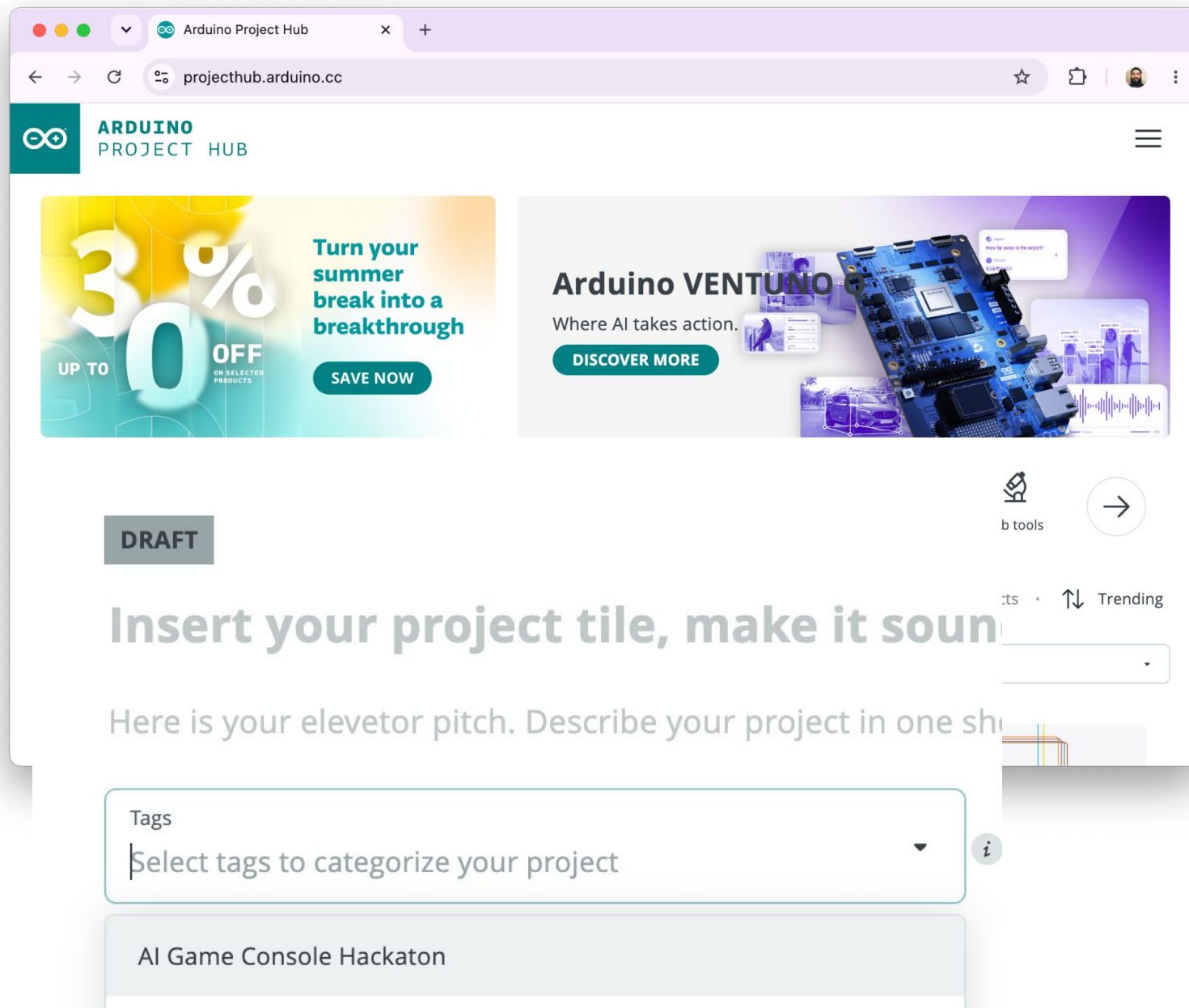
After presentation / Wrap-up

- **Put the hardware back in the box**
- **Return the hardware to the table you collected it**



Post your project on Project Hub!

- Post your project on Project Hub
- Select **"AI Game Console Hackathon"** tag





Project Submission!

- Towards the end of the hackathon, we ask you to submit your project here: <https://forms.gle/gXvNVsALP6PBCkpY8>
- Link will be shared later on as well



Start:

summerengine.com/jam/ai-game-console-hackathon

Summer Credits:

Email timofey@summerengine.com with subject "JAM CREDITS" and the email who need credits

Cursor Credits:

<https://cursor-ai-game-console-hackathon-09-2026.teamdee.l.workers.dev>

Arduino App Lab:

<https://www.arduino.cc/en/software/#app-lab-section>

Elevenlabs:

<https://docs.google.com/document/d/1mCh5MtOzBw0aJpurQVUmIVFfPMAHW3MjNemE-LNiMto/>

- **Wi-Fi name:**
BLOXHUB
- **User:**
WIFI@campus.fmg
- **Password:**
529959

k.soderby@arduino.cc

